Optics Communications 560 (2024) 130448

Contents lists available at ScienceDirect

Optics Communications

journal homepage: www.elsevier.com/locate/optcom

A hybrid MLP-CNN model based on positional encoding for daytime radiative cooler

Xueyu Wang [a], Shuo Chen [a], Lei Chen [a], Danfeng Zhu [b], Yumin Liu [a,*], Tiesheng Wu [c,**]

[a] Beijing University of Posts and Telecommunications, State Key Laboratory of Information Photonics and Optical Communications, No.10, Xitucheng Road, Beijing, 100876, China

[b] School of Physics and Electrical Engineering, Jiaying University, Meizhou, 514015, China

[c] Guilin University of Electronic Technology, College of Information and Communication Engineering, 1 Jinji Road, Guilin, 541004, China

ARTICLE INFO

Keywords:
Positional encoding
Radiative cooling
Multilayer perceptron
Convolutional neural network

ABSTRACT

Radiative cooling is a groundbreaking technology for sustainable temperature regulation to combat global warming and the urban heat island effect. Traditional optimization algorithms for cooling system structures are often brute-force and even limited to local optimal solutions. Here, we propose a joint simulation method based on a multilayer perceptron (MLP) and a convolutional neural network (CNN) with position encoding (PE), namely the hybrid MLP-CNN model with PE. This method can significantly reduce simulation time, avoid getting trapped in local optima, and accurately predict the optical response and radiative cooling power of the structure. Applying PE enables the MLP model to learn the relationship between structural parameters and optical response more quickly and effectively, reducing loss during training and more minor root mean square error (RMSE) and mean relative error (MRE) during testing. In addition, the residual network can reduce the problem of overfitting in the network. Compared to using only an MLP, the hybrid MLP-CNN with PE model can effectively reduce loss and achieve more accurate prediction outputs. The hybrid model is much faster than traditional electromagnetic simulations, taking only 1% of the time, and the model dramatically reduces the time needed to predict the optical response of structures and effectively decreases the cost of designing optical devices. Utilizing this hybrid model not only avoids the potential drawbacks of being limited to suboptimal solutions and significantly reduces the simulation time, but also effectively predicts the optical response of the device. This approach can demonstrate remarkable adaptability in optimizing geometric parameters of various optical apparatuses, including polarization converters, beam splitters, metalens, and other two-dimensional nanostructures.

1. Introduction

In recent decades, the issues of global warming and the energy crisis have garnered significant attention due to the extensive combustion of fossil fuels. Energy conservation has emerged as a prominent research area. Currently, 40%–50% of the total global energy consumption comes from building energy consumption, mainly for lighting, ventilation, heating, and air conditioning [1,2]. The energy demand for cooling has surged to approximately 15% of global energy consumption [3,4]. Consequently, thermal management of buildings has garnered attention in scientific literature. Daytime radiative cooling systems that do not require any input power would greatly alleviate the global energy crisis. Daytime radiative cooling technology leverages the atmospheric transparency window, which spans from 8 to 13 μm, to actively dissipate heat from the Earth into the frigid environment of outer space (3 K). Daytime radiative coolers can achieve passive cooling and maintain temperatures lower than the ambient, even in direct sunlight. They have potential applications in cooling solar cells [5] and air conditioning buildings [6], among other possibilities. During the last decade, researchers have investigated various structures of radiative coolers, including micro pyramid structures [7,8], hole structures [9], polymer structures [10], and multilayer thin-film structures [11,12]. The investigation of multi-layer thin film structures has been the subject of extensive research, primarily due to their uncomplicated composition and straightforward manufacturing process. Chen et al. presented a multilayer thin-film radiation cooler made of Si_3N_4–Si–Al-Substrate to

* Corresponding author.
** Corresponding author.
E-mail addresses: microliuyumin@hotmail.com (Y. Liu), tieshengw@guet.edu.cn (T. Wu).

https://doi.org/10.1016/j.optcom.2024.130448
Received 31 October 2023; Received in revised form 13 February 2024; Accepted 4 March 2024
Available online 12 March 2024

achieve selective thermal radiation in the atmospheric window band. The elimination of parasitic thermal loads is employed to achieve significant reductions in temperature, reaching up to 60 °C below ambient temperature [13]. Kim and colleagues developed a radiative cooler with a multilayer thin-film structure, which operates through an endothermic reaction involving NH_4NO_3/H_2O and driven by water sorption [12]. To optimize the performance of multilayer passive radiative coolers, it is crucial to achieve a broad range of impedance matching within the atmospheric window while simultaneously preventing the absorption of solar radiation within the solar wavelength range. This complex objective requires a thorough strategy that includes conducting extensive experimental studies [14], meticulous optical simulations, and employing classical algorithms to optimize each layer's thickness. Algorithms of this nature often employ a brute-force methodology and are prone to being trapped in local optima. The radiative cooler developed by Wu et al. demonstrates a high level of efficiency in radiative cooling [7]. However, there is potential for further improvement in the radiative cooling efficiency and simplification of the structure through the optimization of the thickness of the multi-layer thin film. The full dielectric quasi-three-dimensional subwavelength structure proposed by He et al. achieves perfect anomalous reflection at optical frequencies through equations and theoretical derivations [15]. However, it is regrettable that further optimization of the structure dimensions to improve device efficiency and simplify the structure was not performed. In the case of other types of two-dimensional optical devices, such as polarization converters [16], beam splitters [17], and metalens [18], the process of calculating the model structure alone can be time-consuming, often requiring several tens of minutes or even hours. Conventional optimization techniques, such as memetic algorithm [19], particle swarm optimization [20], and ant colony optimization [21], are not feasible for optimizing the structural parameters of two-dimensional optical devices.

Recently, the domain of nanophotonics and metamaterial design has experienced integrating deep learning (DL) techniques [22]. This amalgamation has been motivated by the impressive ability of DL to expedite computationally intensive simulation procedures while simultaneously providing flexible design frameworks for a wide array of applications. This approach enables precise engineering of optical properties in various nano photon structures. Researchers have utilized a wide array of well-established neural networks (NN) to advance the boundaries of metamaterial design. These include but are not limited to, multilayer perceptron (MLP), convolutional neural networks (CNNs), recurrent neural networks (RNNs), generative adversarial networks (GANs), radial basis function (RBF), group method of data handling (GMDH) and variational autoencoders. The MLP [23,24], characterized by its simple architecture and ease of implementation, is suitable for handling classification and regression tasks. However, its performance may be limited when dealing with complex data structures such as images and sequential data. CNNs [25] are well-suited for image processing, effectively extracting local features and exhibiting robustness to translations, scalings, and other transformations. Nevertheless, they entail high computational complexity and impose specific requirements on the shape and size of input data. RNNs [26] are adept at handling sequential data, and capturing temporal dependencies, yet they may encounter issues like gradient vanishing or exploding during training. GANs [27] excel in generating highly realistic samples but are susceptible to instability during the training process and prone to mode collapse. RBF networks [28], characterized by their local approximation capability and fast convergence, are sensitive to the selection of centers and widths and are less adept at handling high-dimensional data. GMDH [29] presents a self-organizing approach to model structure construction, suitable for modeling and prediction in complex systems, albeit with relatively low efficiency when dealing with large-scale data. The MLP and CNN are two of the most fundamental and widely applied neural network architectures, and they have also found extensive application in the field of micro-nano optical devices. So et al. introduced a fully connected neural network to predict target spectra accurately. This was achieved by identifying the inherent connection between grating structural parameters and reflectance spectra [30]. Yeung et al. utilized CNN for training to predict the electromagnetic response of metal-dielectric-metal metamaterials, enabling physical discoveries and design optimizations in optics and photonics [31]. Chen et al. designed a neural network based on the Transformer model, which divides the studied spectrum into multiple modules, overcoming the severe overfitting issue of traditional deep learning and enhancing the learning capability [32]. In comparison to traditional numerical methods, these techniques provide a notable decrease in computational demands by directly incorporating the connections between structure and optical response, thereby eliminating the necessity of solving the complex three-dimensional vector Maxwell's equations. The integration of advantages from different networks holds the potential to enhance the learning capability of the network, thereby improving its predictive performance.

Here, we report a method that integrates PE (position encoding) with MLP and CNN to achieve precise prediction of the emission spectrum exhibited by the daytime radiative cooler. Compared to using only MLP or CNN, the hybrid model improves the network's understanding of the complex relationship between structural parameters and optical response. Furthermore, the integration of residual network effectively addresses the problem of overfitting. By integrating PE, the hybrid MLP-CNN with PE model quickly understands the inherent connections between the input and output variables, decreasing the RMSE and MRE. This method, which utilizes PE and integrates MLP and CNN, has demonstrated notable efficacy in optical response. It substantially diminishes simulation duration and circumvents the issue of optimization becoming trapped in local optima. This efficient technique for optimizing structural parameters exhibits promising applicability to various optical devices, including polarization converters, beam splitters, metalens, and other two-dimensional nanostructures.

2. Geometry and methods

Fig. 1(a) depicts the multilayer passive radiative cooler (PRC). The cooling system comprises a hierarchical arrangement of materials, including MgF_2, TiO_2, MgF_2, SiN, SiO_2, MgF_2, and Ag. Ag is used as the substrate because of its high impedance in the solar spectrum, which reduces light absorption. The thickness of each layer in the structure, from top to bottom, is d_1, d_2, d_3, d_4, d_5, and d_6. The refractive indices of all the materials were obtained from previously published literature [33]. The red curve in Fig. 1(b) represents the ideal emission spectrum of the radiative cooler, while the black curve represents the predicted emission spectrum obtained through the hybrid MLP-CNN with PE model. The objective of this work is twofold: to minimize emission within the wavelength range of 0.3–3 μm and to maximize emission within the wavelength range of 8–13 μm. The transfer matrix method (TMM) [34–36] calculates the emission spectrum of the radiative cooler. In the wavelength range spanning from 0.3 to 20 μm, the emission spectrum consists of 395 wavelength points, with a step size of 0.05 μm. Within these six layers (d_1, d_2, d_3, d_4, d_5, and d_6) of the structure, the range of variation for each layer's thickness exhibits a range of 0.5–3 μm, incrementing by 0.5 μm. This approach yields 46,656 emission spectra, comprising 395 data points in each spectrum.

The assessment of radiative coolers' effectiveness relies heavily on the metric of radiative cooling power, and attaining a substantial power output is a primary goal within this field. The cooling power per unit area, represented as P_{cool} (T), for the PRC at temperature T can be expressed as:

$$P_{cool}(T) = P_{rad}(T) - P_{solar} - P_{atm}(T_{amb}) - P_{cc}(T) \tag{1}$$

where the T_{amb} representing the ambient air temperature, is assumed to be 300 K. The power radiated per unit area by the device, denoted as

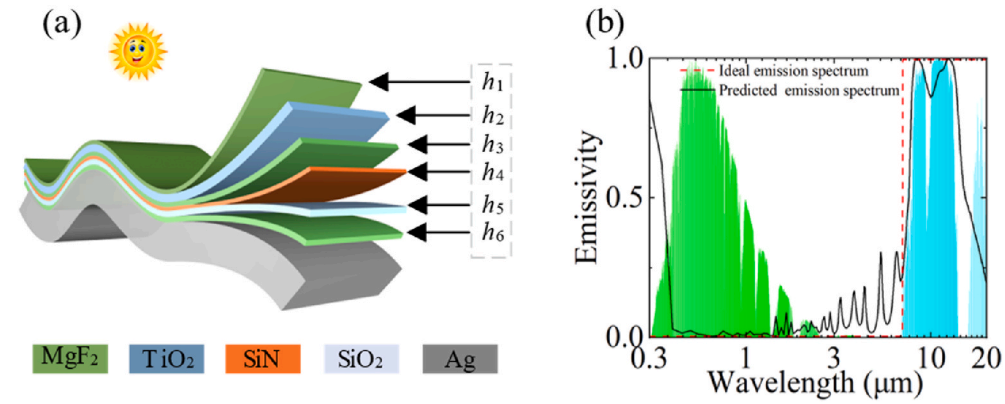

Fig. 1. (a) The schematic diagram of the designed multilayer thin film radiative cooler. (b) The emission spectrum of the ideal radiative cooler (red dotted line) and the predicted by the hybrid MLP-CNN with PE model of this work (black solid line).

$P_{rad}(T)$, the absorbed solar intensity P_{solar}, the power absorbed per unit area due to incident atmospheric radiation P_{atm} (T_{amb}), and the power loss per unit area due to convection and conduction P_{cc}(T) are all considered. It is important to note that variations in T_{amb} are also taken into account. The first three terms on the right-hand side of Eq. (1) are dependent on the emissivity of the device. The emissivity of a multilayer thin film structure, denoted as ε (λ, θ, $\mathbf{d}$, $\mathbf{n}(\lambda)$), is the function of the incidence angle θ, the wavelength λ, the thickness of the layers of the multilayer structure $\mathbf{d} = [d_1, d_2, \ldots, d_M]$, and the refractive index of each layer $\mathbf{n}(\lambda) = [n_1(\lambda), n_1(\lambda), \ldots n_M(\lambda)]$. Here, the subscripts in the vectors $\mathbf{d}$ and $\mathbf{n}(\lambda)$ indicate the layer number, and M represents the total number of layers. The expression for $P_{rad}(T)$ is given by:

$$P_{rad}(T) = \int_{\Omega} \cos\theta \int_0^{\infty} I_{BB}(\lambda, T)\varepsilon(\lambda, \theta, \mathbf{d}, \mathbf{n}(\lambda))\mathrm{d}\lambda\mathrm{d}\Omega \tag{2}$$

where $\int_{\Omega}(\cdot)\mathrm{d}\Omega = 2\pi \int_0^{\pi/2} (\cdot)\sin\theta\, \mathrm{d}\theta$ represents the solid angle integration over a hemisphere. The I_{BB} (λ, T) denotes the emission intensity spectrum of a blackbody, which is determined by Planck's law:

$$I_{BB}(\lambda, T) = \frac{2hc^2}{\lambda^5}\frac{1}{e^{h_c/\lambda k_B T} - 1} \tag{3}$$

where h is Planck's constant, k_B is the Boltzmann constant, and c is the speed of light.

$$P_{solar} = \int_0^{\infty} I_{AM1.5}(\lambda)\varepsilon(\lambda, 0, \mathbf{d}, \mathbf{n}(\lambda))\mathrm{d}\lambda \tag{4}$$

$$P_{atm}(T_{amb}) = \int_{\Omega} \cos\theta \int_0^{\infty} IBB(\lambda, T_{amb})\varepsilon(\lambda, \theta, \mathbf{d}, \mathbf{n}(\lambda))\varepsilon_{atm}(\lambda, \theta)\mathrm{d}\lambda d\Omega \tag{5}$$

$$P_{cc}(T) = h_c(T_{amb} - T) \tag{6}$$

where the $I_{AM1.5}$ represents the AM1.5 solar spectrum with an irradiance of 1000 W m^{-2}. The term ε_{atm} (λ, T) corresponds to the atmospheric emissivity, while h_c denotes the non-radiative heat coefficient, which encompasses the convection heat transfer coefficient. For planar structures subjected to natural convection in air, the convection heat transfer coefficient typically falls within the range of 2–25 W m^{-2} K^{-1}. However, this value can be higher in cases involving forced convection. In this investigation, we consider h_c to be 6.9 W m^{-2} K^{-1}. Furthermore, the atmospheric emissivity is defined as follows:

$$\varepsilon_{atm}(\lambda, \theta) = 1 - t_{atm}(\lambda)^{1/\cos\theta} \tag{7}$$

where t_{atm} is the atmospheric transmittance at the zenith direction, is determined by utilizing the data provided in Ref. [30]. In the present investigation, we employ the transmission matrix method to compute the PRC's reflectivity, $R(\lambda,\theta,\mathbf{d},\mathbf{n}(\lambda)) = 0.5\cdot\{R_{TE}(\lambda,\theta,\mathbf{d},\mathbf{n}(\lambda)) + R_{TM}(\lambda,\theta,\mathbf{d},\mathbf{n}(\lambda))\}$, which is averaged over the incident polarization of light, considering both transverse electric (TE) and transverse magnetic (TM) modes. The emissivity can be calculated by Refs. [37,38]:

$$\varepsilon(\lambda, \theta, \mathbf{d}, \mathbf{n}(\lambda)) = 1 - R(\lambda, \theta, \mathbf{d}, \mathbf{n}(\lambda)) \tag{8}$$

In the absence of any transmission, the right-hand side of Eq. (8) can be attributed to the absorptivity. As per Kirchhoff's law of thermal radiation, in a state of thermodynamic equilibrium, the absorptivity is equivalent to the emissivity [38].

3. Results

3.1. Multilayer perceptron model

The multilayer perceptron (MLP) represents a sophisticated artificial neural network structure characterized by a hierarchical arrangement of interconnected nodes or neurons. The MLP has shown remarkable effectiveness and has been successful in various fields, including image recognition [39], natural language processing [40], and financial forecasting [41], and the prominent role of this tool in artificial intelligence is due to its versatility and ability to adapt. Fig. 2(a) depicts the schematic diagram of our designed multilayer structure of PRC. In Fig. 2(b), the MLP takes d_1, d_2, d_3, d_4, d_5, and d_6 as input data, yielding the emission spectrum of the device as output. The multilayer network we have devised is primarily composed of three distinct components: the input vector, the hidden layers, and the output vector (represented by 3 Gy rectangular boxes in order from top to bottom in Fig. 2(b)). In the input vector section, it is first ensured that the input data (d_1, d_2, d_3, d_4, d_5, and d_6) are within the range of 0–1, which requires dividing them by 5 μm. This normalization process ensures the input data is appropriate for further analysis and calculations. In the hidden layer section: The golden rectangular box represents a linear transformation operation, where Linear (dim$_{input}$, W) denotes the mapping of the input feature vector from a dimension of dim$_{input}$ to W. The light blue color indicates the presence of a non-linear activation function. The W is assigned a value of 256 in the MLP model. The dashed line segment signifies the incorporation of residual connection. By directly propagating the input information to subsequent layers, these residual connection effectively facilitate the propagation of gradients, thereby addressing the challenges of gradient vanishing and model degradation that arise during the training of deep networks [42,43]. The integration of skip connections enhances both the efficacy of training and the expressive capacity of the network. In the output section, the dim$_{output}$ corresponds to the number

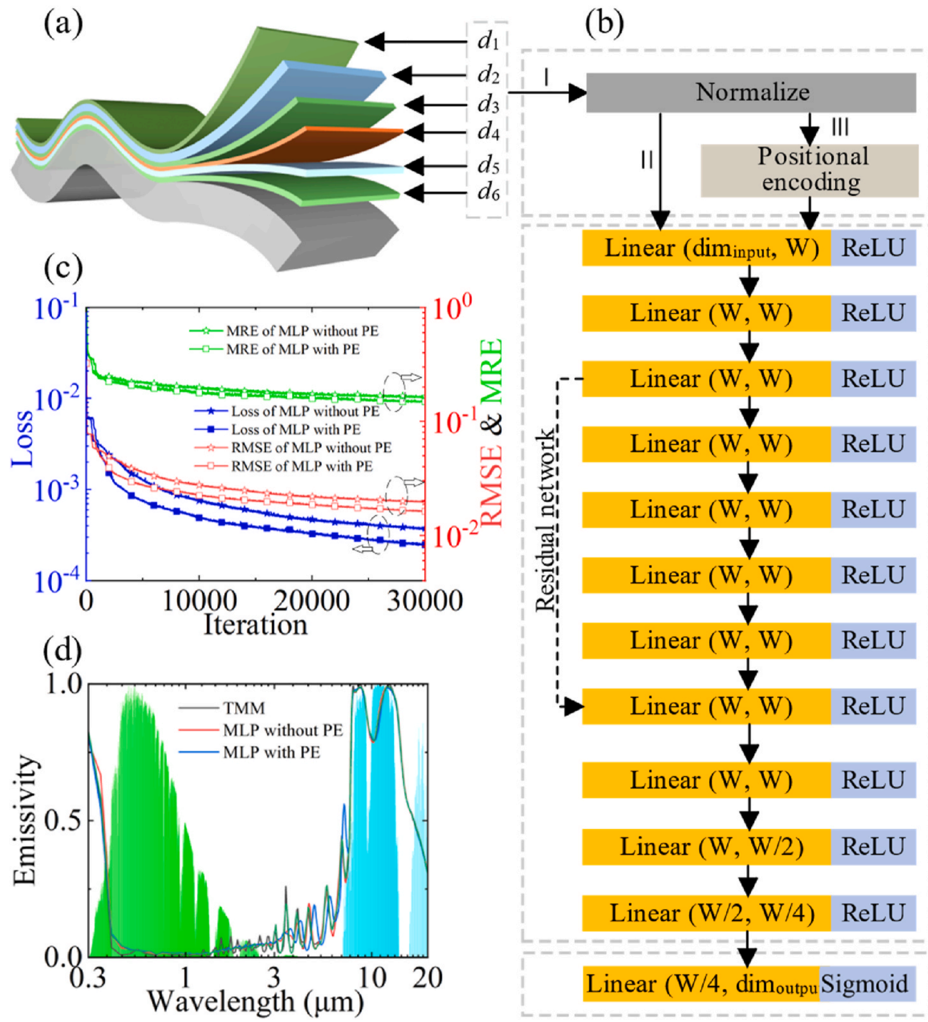

Fig. 2. (a) The multilayer thin film PRC schematic diagram. (b) The visualization of MLP architecture. The data flows through channels I and II, representing an MLP without PE, while the data flows through channels I and III, representing an MLP based on PE. (c) RMSE, MRE, and loss values after 30,000 iterations for MLP with PE and without PE. (d) Predicted emission spectrum of MLP without PE, MLP with PE, and computed emission spectrum through TMM.

of sampling points in the emission spectrum, which is 395. The 46,656 data points calculated by TMM, each containing structural thickness and spectral response, were shuffled randomly and then divided into a training set, which accounted for 90% of the total data, and a validation set, which accounted for 10%. The Loss function utilizes the mean squared error (MSE) to measure the difference between the predicted outcomes and the actual values obtained from optical simulations. The mathematical representation of loss is shown: Loss $= \sum_{i=1}^{N}(y_i - \widehat{y}_i)^2 / N$. The N represents the total number of samples, y_i denotes the actual values obtained from electromagnetic simulations, and $\widehat{y}_i$ signifies the predicted values generated by the model. The RMSE (root mean squared error) and MRE (mean relative error) are widely adopted metric for quantifying the overall disparity between predicted and actual values, thus serving as a reliable indicator of model performance. The expressions for RMSE and MRE are given by RMSE $= \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \widehat{y}_i)^2}$ and MRE $= \frac{1}{N}\sum_{i=1}^{N}\left|\frac{y_i - \widehat{y}_i}{y_i}\right|$. The lower values of RMSE and MRE indicate that the model's predictions are more consistent with electromagnetic simulation results, which is the goal pursued in this work. It is worth noting that Loss is used to measure the model's evaluation on the training set, while RMSE and MRE are used to gauge the model's evaluation on the validation set. A small Loss does not necessarily mean a small RMSE, and the size of the Loss alone cannot explain the accuracy of the model's predictions. Only when the Loss and RMSE are small can we conclude that the model has been sufficiently trained and possesses good predictive capabilities. The Adam optimizer [44] combines the stochastic gradient descent algorithm (SGD) with adaptive learning rate algorithms, enabling fast convergence and reduced training time, and is widely used in deep learning. The networks, including those designed below, all use the Adam optimizer for gradient descent. For all the networks involved in this work, a fixed value for the learning rate ($\text{rate}_{\text{learning}}$) has not been set. Instead, it is a value that changes with the increase in the number of iterations. The $\text{rate}_{\text{learning}}$ is defined: $\text{rate}_{\text{learning}} = 5 \times 10^{-4} \times 0.1^{n_{\text{iteration}}/250000}$, and the $n_{\text{iteration}}$ represents the current number of iterations. Setting such a learning rate allows for a larger learning rate in the early stages of model training, aiding the model in quickly approaching a better solution and expediting the convergence process. In the later stages of model training, this helps the model finely adjust the weights as it approaches the optimal solution, avoiding oscillations near the optimal solution caused by an excessively large learning rate.

3.2. Multilayer perceptron model with positional encoding

In the Transformer model, a commonly adopted strategy for incorporating positional information uses sine and cosine functions for positional encoding [45]. By applying these functions to each element within the positional encoding matrix and subsequently transforming the individual data points into multiple sine and cosine values, it becomes possible to represent each position as a vector of higher dimensionality, as demonstrated in Fig. 2(b) when data passes through channels I and III. Employing positional encoding engenders the extraction of an augmented informational reservoir from a restricted ensemble of input dimensions, thereby facilitating accelerated and enhanced learning within the proposed network. The positional encoding equation [46] is defined as follows:

$$\mathrm{Input}(p) = \left(\sin\left(2^{0}\pi p\right), \cos\left(2^{0}\pi p\right), \cdots\cdots, \sin\left(2^{L-1}\pi p\right), \cos\left(2^{L-1}\pi p\right)\right) \tag{9}$$

where the Input(·) represents the normalized input parameters within the [0,1] range. The transformed high-frequency function is positioned on the right-hand side of the equation, with the value of L set to 12. The network exhibits an input dimensionality of 6. Applying Eq. (9), one can ascertain that the dimensionality after the high-frequency function computation amounts to 150. The W is set 256. Fig. 2(c) displays the loss and RMSE values of the MLP model with and without PE after 30,000 iterations. It is essential to highlight that the MLP does not undergo high-frequency positional encoding when the data transmits through channels I and II. The $\mathrm{dim}_{\mathrm{input}}$ is equal to the size of the normalized input vector, which is 6.

The MLP without PE and the MLP with PE in Fig. 2(c) achieved losses of 0.000378 and 0.000247, with RMSE of 0.0197 and 0.0162, and MRE of 0.165 and 0.149, respectively. The utilization of PE within the MLP is readily observable, as it not only facilitates a swift decrease in loss but also yields a lower loss compared to the model that lacks PE, following an equivalent number of iterations. The findings from the training demonstrate that the MLP with PE yields a reduced RMSE and MSE, suggesting that PE effectively enhances the speed and quality of learning device parameters and optical responses. Furthermore, the MLP without and with PE consumed 8.54 min and 8.55 min, respectively, after 30,000 iterations. This finding indicates that the incorporation of PE does not result in additional computational expenses during the execution of the model. Fig. 2(d) illustrates the emission spectra of the PRC computed by the TMM and the models with and without PE. Both models exhibit a remarkable consistency with the actual values obtained from electromagnetic calculation, effectively predicting the optical response of the PRC. While the MLP with PE exhibits a lower RMSE and MRE, suggesting its superior ability to predict the optical response of the device. This finding underscores the effectiveness of using PE in enhancing the predictive performance of the MLP without PE model. Additionally, the Supporting Information provides the impact of the dimensions and numbers of hidden layers on the performance of the MLP with PE.

3.3. Convolutional neural network model

CNN finds extensive application in computer vision, particularly in tasks such as image and video recognition. In recent years, CNN has made notable progress in optical device design. The fundamental principle that forms the basis of CNN lies in the process of extracting significant features from input data through the utilization of convolutional operations. CNN utilizes a series of convolutional and pooling layers to iteratively extract and diminish the spatial dimensions of the underlying features. Eventually, the classification or regression task is accomplished through fully connected layers. In this work, we introduce the CNN diagram, as illustrated in Fig. 3(b). This network transforms the thickness of the daytime radiative cooler depicted in Fig. 3(a) into a standardized

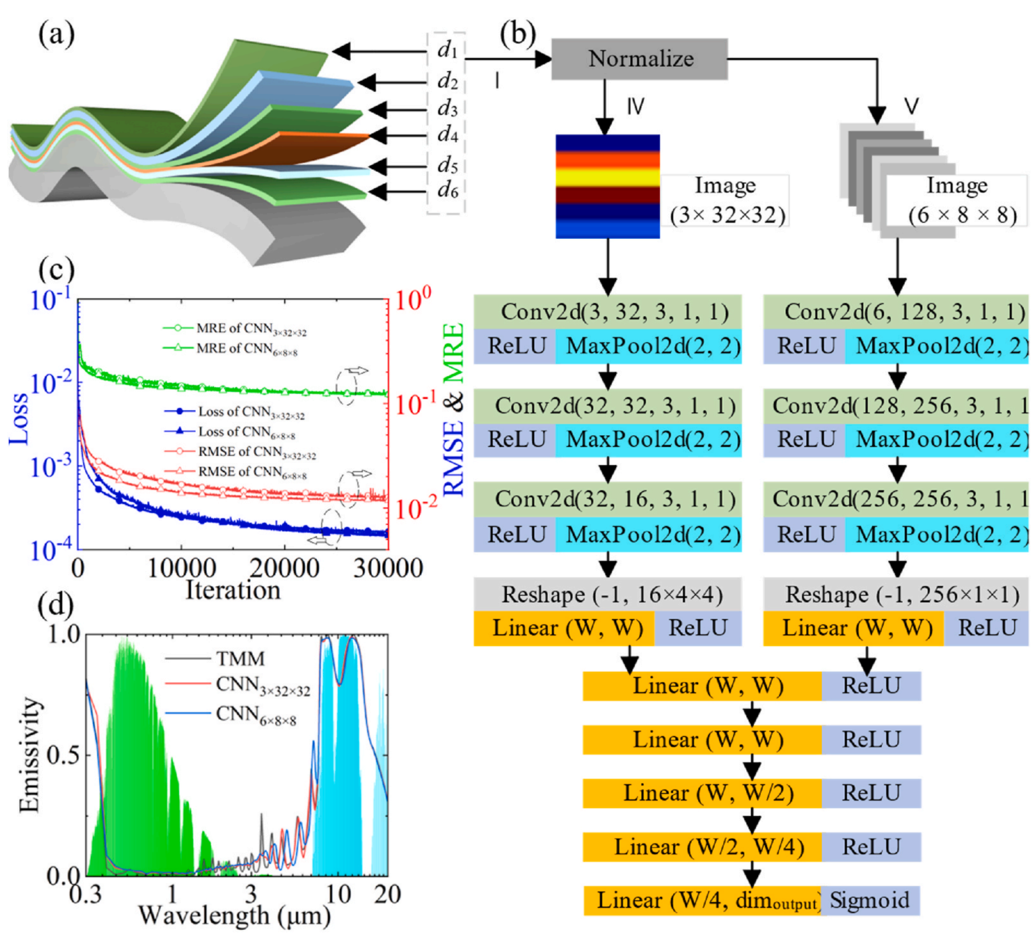

Fig. 3. (a) The schematic diagram of the multilayer thin film PRC. (b) The CNN architecture diagram. (c) The relationship between loss, RMSE, and MRE of $\mathrm{CNN}_{3\times32\times32}$, the $\mathrm{CNN}_{6\times8\times8}$ as a function of iteration. (d) The emission spectra calculated by TMM and predicted by the $\mathrm{CNN}_{3\times32\times32}$, $\mathrm{CNN}_{6\times8\times8}$ model.

value that ranges from 0 to 1. Scheme 1 entails the transformation of normalized thickness measurements into $3 \times 32 \times 32$ image array, and the transformation process is explained in the Supporting Information (The process of transforming a 6×1 vector to a $3 \times 32 \times 32$). This image comprises six rectangular elements, each possessing identical width and length. The rectangular shapes in the image display a range of colors that correspond to the normalized thickness values, indicating different magnitudes. Hence, the information is transformed into a $3 \times 32 \times 32$ tensor (3 representing the red, green, and blue channels). Then a convolution operation is performed on the image, referred to as the $CNN_{3\times32\times32}$ model (data is propagated through channels I and IV in Fig. 3(b)). An alternative approach involves transforming the normalized data into six 8×8 images (the transformation process is explained in the Supporting Information), which can then be subjected to subsequent convolution operations, and the data passes through channels I and V in Fig. 3(b), referred to as the $CNN_{6\times8\times8}$ model. The subsequent part of the CNN is mainly composed of two parts in Fig. 3(b): the convolutional segment and the linear segment. The convolutional segment comprises three convolutional layers, three non-linear activation functions (ReLU), and three max-pooling layers. The linear segment encompasses four linear layers accompanied by a non-linear activation function. The Conv2d (6, 128, 3, 1, 1) convolutional layer encompasses a set of five numerical parameters enclosed within parentheses, denoting the input channels, output channels, kernel size, stride, and padding, respectively. MaxPool2d (2, 2) refers to a two-dimensional max-pooling operation that partitions the input into non-overlapping regions of size 2×2. Within each region, the operation selects the maximum value as the output. This process effectively reduces the spatial dimensions of the input feature map by half in both height and width, facilitating subsequent processing.

After 10,000 iterations, the designed $CNN_{3\times32\times32}$ and $CNN_{6\times8\times8}$ achieved a loss of 0.000165 and 0.000147, and the RMSE of 0.0127, and 0.0119, and the MRE of 0.122 and 0.122, respectively as illustrated in Fig. 3(c). The entire training process of $CNN_{3\times32\times32}$ and $CNN_{6\times8\times8}$ consumed 6.7 and 6.2 h. Compared to the MLP with PE model proposed in Fig. 2(b), the $CNN_{3\times32\times32}$ and $CNN_{6\times8\times8}$ models require more time at the same number of iterations but exhibit smaller loss and RMSE. The CNN model can better learn the relationship between structural parameters and output responses. Compared to $CNN_{3\times32\times32}$, the $CNN_{6\times8\times8}$ exhibits smaller Loss and RMSE, demonstrating better performance in predicting the optical response of structures. In the following content, if not explicitly mentioned, the CNN model refers to the $CNN_{6\times8\times8}$ model. In Fig. 3(d), the emission spectra obtained through the use of the TMM, predicted by the $CNN_{3\times32\times32}$ and the $CNN_{6\times8\times8}$ model are represented by the black, red, and blue curves, respectively. Evidently, for multilayer thin film structures, the CNN network also demonstrates commendable performance in optical response prediction.

3.4. The hybrid multilayer perceptron-convolutional neural network model

The MLP model cannot leverage the local structure and spatial information within the data, rendering its performance potentially inferior to that of the CNN model in the case of spatially related data. Additionally, the MLP model's larger parameter count poses a risk of overfitting, necessitating the utilization of more training data and regularization techniques to mitigate this issue. The CNN model exhibits a heightened sensitivity towards the dimensions and shapes of input data, necessitating preprocessing and adjustments. Additionally, when faced with non-image data, the performance of CNN models may not exceed that of the MLP. Here, we propose a model that integrates the merits of MLP and CNN, effectively leveraging the benefits of each approach to their maximum capacity. Through convolution and pooling operations employed in CNN, the model demonstrates its ability to extract spatial features from the input data. Subsequently, through the utilization of fully connected layers in MLP, the model can obtain sophisticated representations derived from these features. This allows for prediction and improves the overall performance and generalization ability of the model [47,48]. The incorporation of PE facilitates the acceleration and improvement of the model's acquisition of the correlation between the input and output. The proposed model architecture is illustrated in Fig. 4(b). After standardizing the thickness data of the PRC, it is partitioned into two segments to facilitate subsequent analysis. One segment is subjected to convolutional operations, which convert the data into six 8×8 images. These images then undergo further convolution, pooling, and flattening operations. Fig. 4(c) illustrates the loss, RMSE and MRE of the MLP with PE, CNN, and hybrid MLP-CNN with PE models following 30,000 iterations. After 30,000 iterations, the MLP with PE, CNN, and hybrid MLP-CNN with PE models achieved losses of 0.000247, 0.000147, and 0.000105, respectively. Additionally, the RMSE for these models was found to be 0.0162, 0.0119, and 0.0119, and the MRE for these three models are 0.149, 0.122, and 0.122. The hybrid MLP-CNN with PE model has the smallest loss among the three models, demonstrating better learning of the relationship between structural dimensions and optical response. Under the same number of iterations, the hybrid MLP-CNN with PE model exhibits a faster and more substantial decrease in Loss, which is attributed to the integrated advantages of MLP and CNN models. Additionally, the presence of positional encoding contributes to the hybrid model's ability to extract more structural information. Fig. 4(d) presents the emission spectra of PRC computed by the TMM and predicted by the MLP with PE, CNN, and hybrid MLP-CNN with PE models. Both the hybrid model and the CNN model exhibit excellent consistency with the electromagnetic simulation results in predicting spectral values. Compared to the MLP model, the hybrid model and CNN model required more time; nevertheless, they demonstrated lower RMSE and exhibited superior accuracy in predicting the optical response of PRC. To visually illustrate the performance of five different artificial neural networks in training emission spectra, Table 1 provides the training time and Loss on the training set (radiation spectra with 41,990 samples) after 30,000 iterations, as well as the testing time, RMSE and MRE on the test set (4666 samples). It is worth noting that, after randomly shuffling 46,656 samples, 90% (41,990 samples) are used for training, and 10% (4666 samples) are used for testing. Table 1 also shows the time required to calculate 4,826,809 samples using the TMM method for comparison. In terms of simulation time, when dealing with a large amount of data (e.g., 4,826,809 samples), both the hybrid model and CNN indeed require more computational time compared to the TMM. However, it is worth noting that for the MLP and MLP with PE models, they are capable of significantly reducing computation time, thereby improving simulation efficiency. It is important to emphasize that TMM computation for one data point typically takes a few milliseconds, whereas for a two-dimensional micro-nano optical structure, it sometimes requires several tens of minutes. For two-dimensional micro-nano structures, using deep neural network models can further save time, which is of great significance in practical applications.

3.5. Prediction of radiative cooling power

Fig. 5(a) illustrates the Loss, RMSE, and MRE of the MLP with PE, CNN, and hybrid MLP-CNN with PE models in predicting radiative cooling power. It can be observed that the hybrid MLP-CNN model exhibits a faster convergence and lower loss, RMSE, and MRE compared to the MLP with PE and CNN models. It is worth noting that to account for the significantly large magnitudes of radiative cooling power (exceeding 1), a normalization procedure is employed by dividing all values by 150 W/m^2. The dim_{output} in the three models is set to 1. These findings further validate the efficacy of the proposed position-encoded hybrid MLP-CNN with PE model in this work. During the training process, the thickness increment of the PRC is set to 0.5 μm, resulting in a dataset of 46,656 samples. For prediction, a step size of 0.2 μm was employed, generating a dataset of 4,826,809 samples. In the training and prediction process, the model's step size is set to 0.5 and 0.2 μm, respectively, and

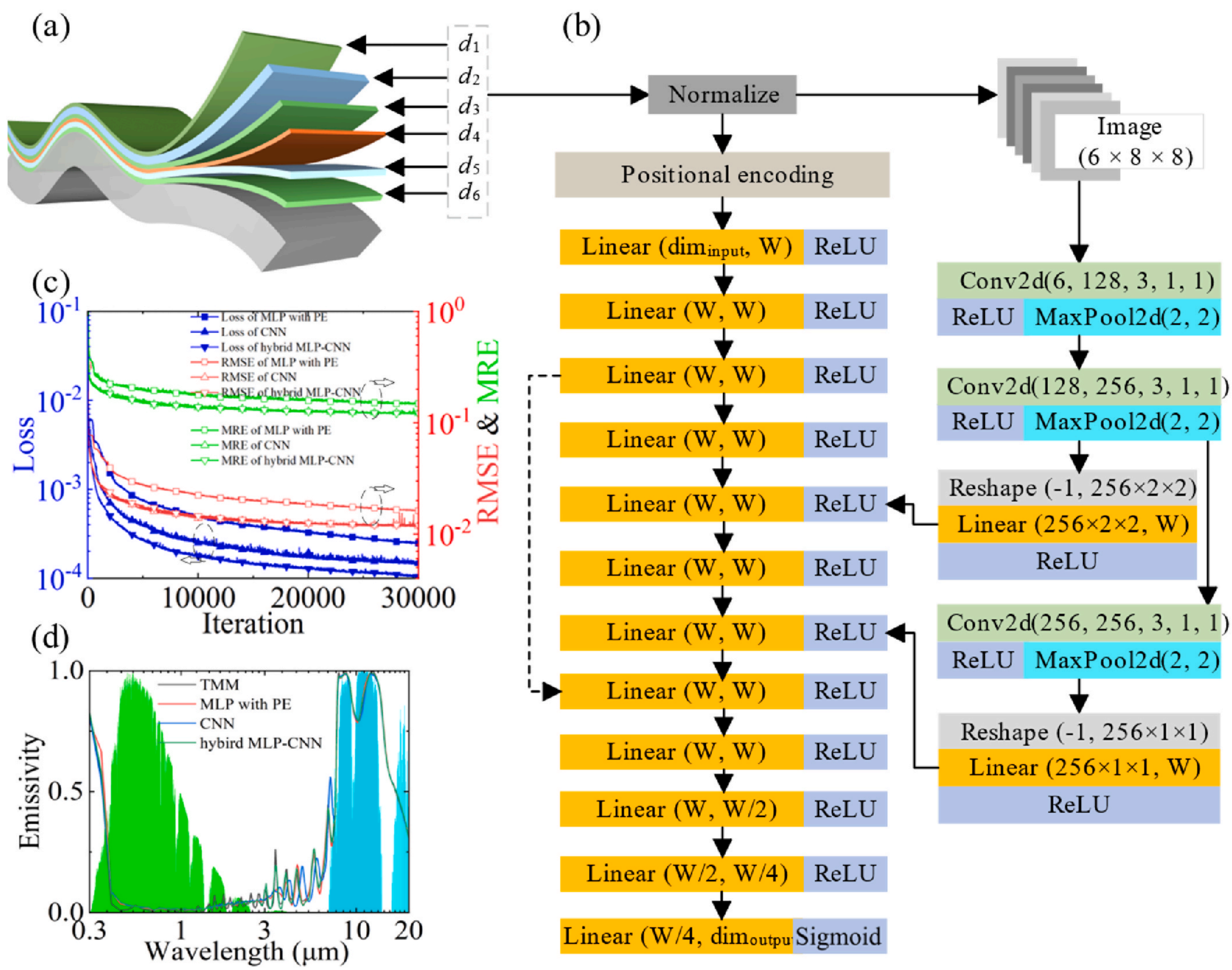

Fig. 4. (a) The multilayer thin film PRC is illustrated in the schematic diagram. (b) The network schematic depicting the collaborative integration of MLP and CNN. W = 256. (c) The relationship between the loss, RMSE and MRE of the MLP with PE, CNN, and hybrid MLP-CNN with PE models to the number of iterations. (d) The emission spectra predicted by the MLP with PE, CNN, and hybrid MLP-CNN with PE models and the emission spectra computed by TMM.

Table 1
The training time, Loss of five different artificial neural networks after 30,000 iterations of training radiation spectra with 41,990 samples, and the testing time, RMSE and MRE in the test set with 4666 samples. And the time consumed for the calculation of 4,826,809 samples using the TMM method for comparison.

Method	Training time of 41,990 samples	Predicting time of 4,826,809 samples	Training Loss of 4666 samples	Testing RMSE of 4666 samples	Testing MRE of 4666 samples
MLP	8.5 min	22 s	0.000378	0.0197	0.165
MLP with PE	8.5 min	22 s	0.000247	0.0162	0.149
$CNN_{3\times32\times32}$	7.2 h	7.5 min	0.000165	0.0127	0.122
$CNN_{6\times8\times8}$	7.2 h	7.5 min	0.000147	0.0119	0.122
Hybrid MLP-CNN with PE	7.3 h	7.7 min	0.000105	0.0119	0.122
TMM	2.8 h (for 4,826,809 samples)		–	–	–

this approach effectively avoids local optimization and maximizes the possibility of achieving the global optimum. Due to the significant reduction in the output data required by the models, the loss decreases rapidly for all three models, as shown in Fig. 5(a). Evidently, among the three models, the hybrid MLP-CNN with PE model demonstrates the best training performance, which aligns with the scenario of predicting emission spectra. Inputting 4,826,809 samples into the hybrid MLP-CNN model, we obtain the radiative cooling power for each data and determine the structural parameters associated with the maximum radiative cooling power. The structural parameters corresponding to the maximum power are considered the best structure for the hybrid MLP-CNN model. Obtaining the optimal structures for the MLP and CNN models follows a similar method as described above. The highest predicted radiative power by the hybrid MLP-CNN with PE model amounts to 113.3 W/m^2, with corresponding structural parameters of 1.9, 0.5, 2.9, 1.7, 1.9, and 1.1 μm. Applying Eqs. (1)–(8) yields an emission power of 113.0 W/m^2, which concurs with the predictions of the hybrid model, thereby validating the efficacy of the hybrid model in predicting the optical response of the device. The emission spectra obtained by applying the TMM to these structural parameters are shown as the red curve in Fig. 5(b), and the radiant cooling power is shown in number 1 in the illustration. After obtaining these thickness parameters, by varying the thickness of each layer by ±0.1 μm, the other lighter-colored curves in Fig. 5(b) can be obtained, and the corresponding radiative cooling power is shown as number 2 to 13 in the insert map, and these modifications do not significantly alter the radiation spectrum or the radiation efficiency. It should be noted that in this investigation, employing a

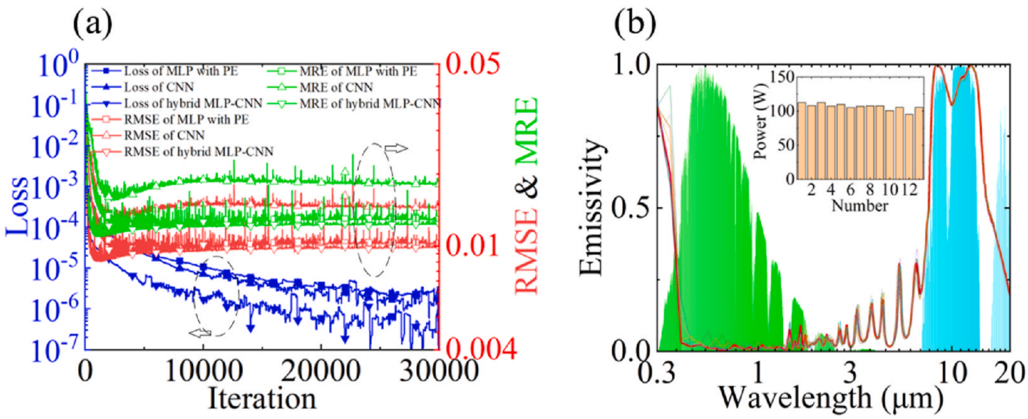

Fig. 5. (a) The loss, RMSE, and MRE of three distinct models in predicting radiative cooling power. (b) The emission spectrum of the structural parameters corresponding to the maximum predicted radiative cooling power is determined using the hybrid MLP-CNN with PE model, represented by the red curve, and the radiative cooling power is shown in number 1 in the illustration. The other lighter-colored curves are the emission spectra obtained by varying the thickness of each layer by ±0.1 μm based on the maximum radiation spectrum. The numbers 2 to 13 in the illustration correspond to the radiative cooling power of these adjusted emission spectra.

step size of 0.2 μm for the prediction model may not yield the most optimal outcomes. Nevertheless, Fig. 5(b) illustrates that this deviation falls within the acceptable range established in this work. It is worth noting that the TMM method would require 835.3 h (see Supporting Information for details) to compute these data, whereas the hybrid MLP-CNN with PE model only takes 7.4 h (7.3 h for training and 7.1 min for prediction). The proposed hybrid MLP-CNN with PE model effectively predicts the optical response of devices while significantly reducing computational time. It is worth noting that if electromagnetic calculations and model simulations are performed separately for two-dimensional micro-nano optical structures, the time comparison becomes even more apparent. To visually demonstrate the performance of three different artificial neural networks in training radiative cooling power, Table 2 provides the training time and Loss on the training set after 30,000 iterations, as well as the testing time, RMSE and MRE on the test set. Table 2 also gives the time required to calculate 4,826,809 samples using the TMM method for comparison. After comparison, although both the hybrid model and CNN model show increased simulation time compared to the MLP with PE model, the hybrid and CNN model demonstrate superior predictive capability. Compared to using the TMM for computation, the hybrid model significantly saves simulation time, greatly alleviating the burden of designing optical structures. This advantage is particularly evident when dealing with complex and large-scale optical structures, making the hybrid model more competitive in practical applications.

Table 2
The training time, training loss of three different artificial neural networks after 30,000 iterations of training radiative cooling power with 41,990 samples, and the testing time, RMSE and MRE in the test set with 4666 samples. And the time required to calculate 4,826,809 samples using the TMM method for comparison.

Method	Training time of 41,990 samples	Predicting time of 4,826,809 samples	Training Loss of 4666 samples	Testing RMSE of 4666 samples	Testing MRE of 4666 samples
MLP with PE	8.5 min	20 s	2.37×10^{-6}	0.0102	0.0127
$CNN_{6\times8\times8}$	7.2 h	7.0 min	1.05×10^{-6}	0.0141	0.0173
Hybrid MLP-CNN with PE	7.3 h	7.1 min	2.31×10^{-7}	0.0099	0.0122
TMM	835.3 h (for 4,826,809 samples)	–	–	–	–

The emissivity dependence on the angle is crucial for cooling performance, and achieving high emissivity over a wide-angle range is the desired objective. Therefore, Fig. 6(a) and (b) also simulate the emissivity under TE-polarization and TM-polarization with incident angles ranging from 0° to 90°. Under TE-polarization, the emissivity of the PRC remains high even at an incident angle of 60°, while under TM polarization, the PRC maintains a good emissivity even at incident angles exceeding 80°. Furthermore, under TM-polarization, as the incident angle exceeds 20°, the emissivity of the PRC gradually increases around a wavelength of 15 μm, which is advantageous for radiative cooling. Fig. 6(c) also simulates the relationship between the average emissivity of the PRC and the incident angle in the solar spectrum (0.3–3 μm) and the atmospheric window spectrum (8–13 μm) under unpolarized conditions. It can be seen that for incident angles less than 50°, the average emissivity of the PRC in the atmospheric window spectrum is greater than 0.9, ensuring the efficiency of radiative cooling. Fig. 6(c) demonstrates that PRC maintains low absorption throughout the entire range of incident angles in the solar spectrum, effectively enhancing its capability for daytime radiative cooling. Based on the aforementioned analysis, it is evident that the proposed PRC exhibits a high emissivity over a wide-angle range.

Fig. 7 displayed the temperature difference between the device and the surrounding environment under different thermal conductivity, considering an ambient temperature of 300 K. The proposed radiative cooler achieves a net cooling power of 113.0 W/m^2 when the ambient temperature equals the temperature of the device. When $h_c = 6.9$ W m^{-2} K^{-1}, the radiative cooler is capable of achieving a remarkable temperature reduction of 11.5 °C compared to the surrounding environment, showcasing its outstanding radiative cooling performance. Taking non-radiative heat exchange into consideration, Fig. 7 considers four combinations of conduction and convection coefficients: $h_c = 0$, 6.9, 12, and 25 W m^{-2} K^{-1}. It is evident that even for $h_c = 25$ W m^{-2} K^{-1}, the proposed cooler still achieves a cooling effect of 4.1 °C lower than the surrounding environment. The above analysis highlights the reliability and performance advantages of the radiative cooler under various environmental conditions, providing robust technical support for addressing issues such as global warming and the urban heat island effect.

4. Discussion and conclusions

Through the analysis and discussion of the loss and RMSE of MLP models with and without PE, it can be inferred that PE enables the conversion of a small amount of input data into more information

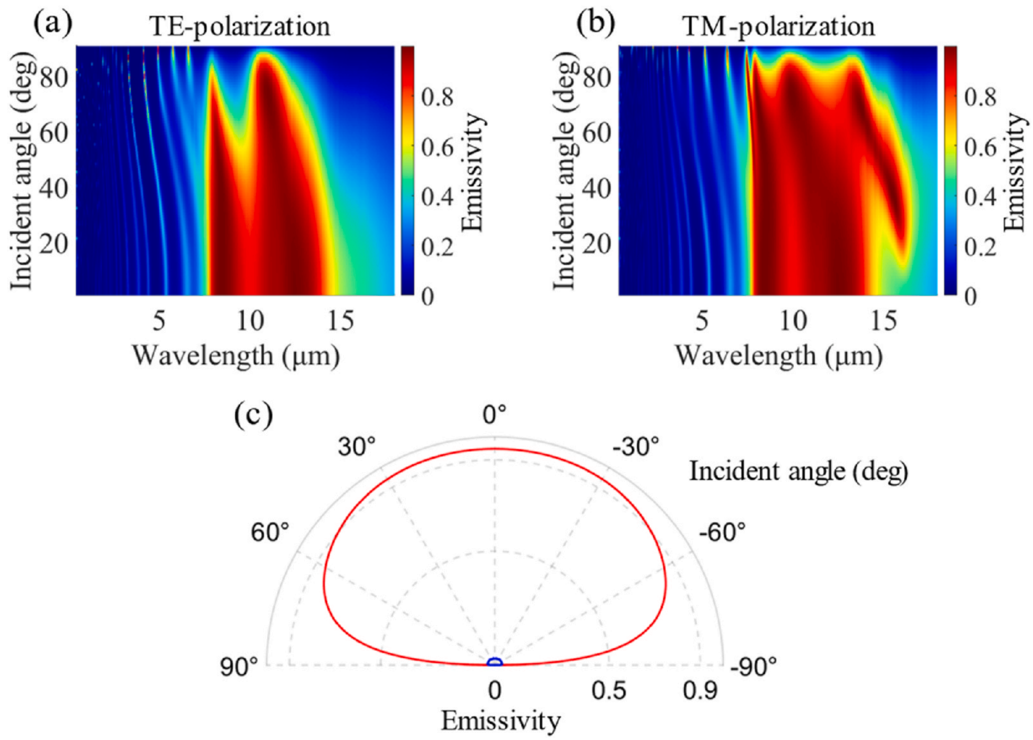

Fig. 6. The emission spectra of the designed PRC vary with different incident angles in (a) TE-polarization and (b) TM-polarization. (c) The designed PRC investigates the relationship between the average emissivity and the incident angle in the wavelength ranges of 0.3–3 μm (blue curve) and 8–13 μm (red curve).

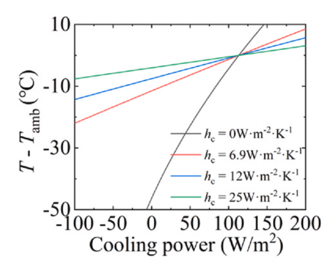

Fig. 7. The impact of different heat coefficients (h_c) and temperature differentials on the radiative power. The ambient temperature (T_{amb}) is set to 300 K.

through high-frequency functions, thereby facilitating the network to learn the relationship between optical response and input parameters more quickly and effectively. The CNN model is also utilized here for training and testing the structural parameters of the input devices, and the one approach is to map the thickness parameter to different colors in an RGB image, where smaller values tend towards green and larger values tend towards red. Another approach is to map the thickness parameter to grayscale images with varying intensities, where larger values correspond to darker shades. Through comparison, the latter approach demonstrates a better prediction of the optical response of the structure. This study combines the advantages of MLP and CNN models and proposes a hybrid MLP-CNN with PE model, which, compared to standalone MLP or CNN models, can better learn the relationship between network input and output, and achieve lower loss and RMSE. By comparing the loss, RMSE, MRE, and simulation/computation time, the exceptional ability of the hybrid model to predict optical response is highlighted. When predicting emission spectra, compared to using TMM calculations, the MLP with PE network saves a significant amount of computational time, while the hybrid model, although taking more time, delivers better predictive results. In designing radiative cooling power, the hybrid model consumes less time compared to the TMM method and can accurately predict the emission cooling efficiency. When designing two-dimensional nanostructures, this hybrid method can save more time compared to traditional electromagnetic calculation methods. The hybrid model demonstrates good prediction of the radiative cooling power of PRC, with a maximum predicted emission power of 113 W/m^2, showcasing its wide-angle absorption characteristics. This hybrid MLP and CNN with PE model not only avoids getting trapped in local optima but also significantly reduces simulation time. It effectively predicts the optical response of devices and demonstrates excellent adaptability in the structural optimization of other optical devices, such as polarization converters, beam splitters, metalens, and other two-dimensional nanostructures.

Funding

Natural Science Foundation of Beijing (2192036); Guangxi Key Laboratory of Wireless Wideband Communication and Signal Processing; The Fundamental Research Funds for the Central Universities (2023RC27).

CRediT authorship contribution statement

Xueyu Wang: Writing – original draft, Writing – review & editing. **Shuo Chen:** Software, Supervision, Validation. **Lei Chen:** Validation, Writing – review & editing. **Danfeng Zhu:** Conceptualization, Validation. **Yumin Liu:** Validation, Writing – original draft, Writing – review & editing. **Tiesheng Wu:** Validation, Writing – review & editing.

Declaration of competing interest

No potential conflict of interest was reported by the authors.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.optcom.2024.130448.

References

[1] Y. Ke, J. Chen, G. Lin, S. Wang, Y. Zhou, J. Yin, P.S. Lee, Y. Long, Smart windows: electro-, thermo-, mechano-, photochromics, and beyond, Adv. Energy Mater. 9 (2019).
[2] U. S. E. I. Administration, "Total Energy Annual Data," https://www.eia.gov/totalenergy/data/annual/.
[3] E.A. Goldstein, A.P. Raman, S. Fan, Sub-ambient non-evaporative fluid cooling with the sky, Nat. Energy 2 (2017) 1–7.
[4] B. Zhu, W. Li, Q. Zhang, D. Li, X. Liu, Y. Wang, N. Xu, Z. Wu, J. Li, X. Li, P. B. Catrysse, W. Xu, S. Fan, J. Zhu, Subambient daytime radiative cooling textile based on nanoprocessed silk, Nat. Nanotechnol. 16 (2021) 1342–1348.
[5] W. Li, Y. Shi, K. Chen, L. Zhu, S. Fan, A comprehensive photonic approach for solar cell cooling, ACS Photonics 4 (2017) 774–782.
[6] Y. Zhai, Y. Ma, S.N. David, D. Zhao, R. Lou, G. Tan, R. Yang, X. Yin, Scalable-manufactured randomized glass-polymer hybrid metamaterial for daytime radiative cooling, Science 355 (2017) 1062–1066.
[7] D. Wu, C. Liu, Z. Xu, Y. Liu, Z. Yu, L. Yu, L. Chen, R. Li, R. Ma, H. Ye, The design of ultra-broadband selective near-perfect absorber based on photonic structures to achieve near-ideal daytime radiative cooling, Mater. Des. 139 (2018) 104–111.
[8] M.M. Hossain, B. Jia, M. Gu, A metamaterial emitter for highly efficient radiative cooling, Adv. Opt. Mater. 3 (2015) 1047–1051.
[9] E. Rephaeli, A. Raman, S. Fan, Ultrabroadband photonic structures to achieve high-performance daytime radiative cooling, Nano Lett. 13 (2013) 1457–1461.
[10] T. Wang, Y. Wu, L. Shi, X. Hu, M. Chen, L. Wu, A structural polymer for highly efficient all-day passive radiative cooling, Nat. Commun. 12 (2021) 365.
[11] A.P. Raman, M.A. Anoma, L. Zhu, E. Rephaeli, S. Fan, Passive radiative cooling below ambient air temperature under direct sunlight, Nature 515 (2014) 540–544.
[12] S. Kim, J.H. Park, J.W. Lee, Y. Kim, Y.T. Kang, Self-recovering passive cooling utilizing endothermic reaction of NH4NO3/H2O driven by water sorption for photovoltaic cell, Nat. Commun. 14 (2023) 2374.
[13] Z. Chen, L. Zhu, A. Raman, S. Fan, Radiative cooling to deep sub-freezing temperatures through a 24-h day–night cycle, Nat. Commun. 7 (2016) 13729.
[14] D. Chae, S. Son, Y. Liu, H. Lim, H. Lee, High-performance daytime radiative cooler and near-ideal selective emitter enabled by transparent sapphire substrate, Adv. Sci. 7 (2020) 2001577.
[15] T. He, T. Liu, S. Xiao, Z. Wei, Z. Wang, L. Zhou, X. Cheng, Perfect anomalous reflectors at optical frequencies, Sci. Adv. 8 (2022) eabk3381.
[16] Y. Sun, Y. Liu, T. Wu, Y. Wang, J. Li, H. Ye, H. Fan, X. Wang, All-dielectric metasurface for linear-polarization conversion with an arbitrary polarization rotating angle, Opt Laser. Technol. 157 (2023) 108762.
[17] J. Li, H. Ye, T. Wu, Y. Liu, Z. Yu, Y. Wang, Y. Sun, L. Yu, Ultra-broadband large-angle beam splitter based on a homogeneous metasurface at visible wavelengths, Opt Express 28 (2020) 32226–32238.
[18] X. Ou, T. Zeng, Y. Zhang, Y. Jiang, Z. Gong, F. Fan, H. Jia, H. Duan, Y. Hu, Tunable polarization-multiplexed achromatic dielectric metalens, Nano Lett. 22 (2022) 10049–10056.
[19] Y. Shi, W. Li, A. Raman, S. Fan, Optimization of multilayer optical films with a memetic algorithm and mixed integer programming, ACS Photonics 5 (2017) 684–691.
[20] D. Chae, M. Kim, P.-H. Jung, S. Son, J. Seo, Y. Liu, B.J. Lee, H. Lee, Spectrally selective inorganic-based multilayer emitter for daytime radiative cooling, ACS Appl. Mater. Interfaces 12 (2020) 8073–8081.
[21] Y. Zhou, W. Li, X. Wang, Y. Qiu, W. Shen, Adaptive gradient descent enabled ant colony optimization for routing problems, Swarm Evol. Comput. 70 (2022) 101046.
[22] D. Mengu, M.S.S. Rahman, Y. Luo, J. Li, O. Kulce, A. Ozcan, At the intersection of optics and deep learning: statistical inference, computing, and inverse design, Adv. Opt. Photonics 14 (2022) 209–290.
[23] J.H. Han, Y.-C. Lim, R.M. Kim, J. Lv, N.H. Cho, H. Kim, S.D. Namgung, S.W. Im, K. T. Nam, Neural-network-Enabled design of a chiral plasmonic nanodimer for target-specific chirality sensing, ACS Nano 17 (2023) 2306–2317.
[24] M. Firouzinia, G. Shafabakhsh, Investigation of the effect of nano-silica on thermal sensitivity of HMA using artificial neural network, Construct. Build. Mater. 170 (2018) 527–536.
[25] R. Masuda, H. Iwabuchi, K.S. Schmidt, A. Damiani, R. Kudo, Retrieval of cloud optical thickness from sky-view camera images using a deep convolutional neural network based on three-dimensional radiative transfer, Rem. Sens. 11 (2019) 1962.
[26] Y. Tang, J. Fan, X. Li, J. Ma, M. Qi, C. Yu, W. Gao, Physics-informed recurrent neural network for time dynamics in optical resonances, Nat. Comput. Sci. 2 (2022) 169–178.
[27] S. An, B. Zheng, H. Tang, M.Y. Shalaginov, L. Zhou, H. Li, M. Kang, K. A. Richardson, T. Gu, J. Hu, Multifunctional metasurface design with a generative adversarial network, Adv. Opt. Mater. 9 (2021) 2001433.
[28] P. Hušek, On monotonic radial basis function networks, IEEE Trans. Cybern. (2022).
[29] M. Amiri, S. Soleimani, ML-based group method of data handling: an improvement on the conventional GMDH, Complex Intell. Syst. 7 (2021) 2949–2960.
[30] S. So, Y. Yang, T. Lee, J. Rho, On-demand design of spectrally sensitive multiband absorbers using an artificial neural network, Photon. Res. 9 (2021) B153–B158.
[31] C. Yeung, J.-M. Tsai, B. King, Y. Kawagoe, D. Ho, M.W. Knight, A.P. Raman, Elucidating the behavior of nanophotonic structures through explainable machine learning algorithms, ACS Photonics 7 (2020) 2309–2318.
[32] W. Chen, Y. Gao, Y. Li, Y. Yan, J.Y. Ou, W. Ma, J. Zhu, Broadband solar metamaterial absorbers empowered by transformer-based deep learning, Adv. Sci. 10 (2023) 2206718.
[33] E.D. Palik, Handbook of Optical Constants of Solids, Academic press, 1998.
[34] A.K. Chowdhary, D. Sikdar, Design of electrotunable all-weather smart windows, Sol. Energy Mater. Sol. Cell. 222 (2021) 110921.
[35] C.C. Katsidis, D.I. Siapkas, General transfer-matrix method for optical multilayer systems with coherent, partially coherent, and incoherent interference, Appl. Opt. 41 (2002) 3978–3987.
[36] H.A. Macleod, Thin-film Optical Filters, CRC press, 2017.
[37] Y. Shi, W. Li, A. Raman, S. Fan, Optimization of multilayer optical films with a memetic algorithm and mixed integer programming, ACS Photonics 5 (2018) 684–691.
[38] H. McMahon, Thermal radiation from partially transparent reflecting bodies, JOSA 40 (1950) 376–380.
[39] R. Janković Babić, A comparison of methods for image classification of cultural heritage using transfer learning for feature extraction. Neural Computing and Applications, 2023.
[40] F. Ghanbari-Adivi, M. Mosleh, Text emotion detection in social networks using a novel ensemble classifier based on Parzen Tree Estimator (TPE), Neural Comput. Appl. 31 (2019) 8971–8983.
[41] D. Wu, X. Ma, D.L. Olson, Financial distress prediction using integrated Z-score and multilayer perceptron neural networks, Decis. Support Syst. 159 (2022) 113814.
[42] K. He, X. Zhang, S. Ren, J. Sun, Identity mappings in deep residual networks, in: Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, pp. 630–645. *Proceedings, Part IV 14* (Springer2016).
[43] G. Huang, Z. Liu, L. Van Der Maaten, K.Q. Weinberger, Densely connected convolutional networks, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 4700–4708.
[44] D.P. Kingma, J. Ba, Adam: A Method for Stochastic Optimization, 2014 arXiv preprint arXiv:1412.6980.
[45] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, Adv. Neural Inf. Process. Syst. 30 (2017).
[46] B. Mildenhall, P.P. Srinivasan, M. Tancik, J.T. Barron, R. Ramamoorthi, R. Ng, Nerf: representing scenes as neural radiance fields for view synthesis, Commun. ACM 65 (2021) 99–106.
[47] O. El Alani, M. Abraim, H. Ghennioui, A. Ghennioui, I. Ikenbi, F.-E. Dahr, Short term solar irradiance forecasting using sky images based on a hybrid CNN–MLP model, Energy Rep. 7 (2021) 888–900.
[48] A. Abbaskhah, H. Sedighi, H. Marvi, Infant cry classification by MFCC feature extraction with MLP and CNN structures, Biomed. Signal Process Control 86 (2023) 105261.